

Computational Reproduction Report

Collective Motion and Inertial Prey–Predator Dynamics

Sneha

Indian Institute of Science Education and Research Kolkata

Supervisor: Dr. Rumi De, IISER Kolkata

June 2026

Abstract

This report presents a computational reproduction of two theoretical studies on collective animal motion and prey–predator dynamics from the group of Dr. Rumi De at IISER Kolkata. The first paper (De and Chakraborty, 2022) reviews self-propelled particle models for collective motion in fish schools, bird flocks, and prey swarms. The second (Chakraborty, Laha, and De, 2022) introduces inertial forces into the prey–predator model and identifies an optimal predator-to-prey mass ratio governing the arms race between pursuit and evasion. All simulations are implemented in Julia (physics kernels) with Python (NumPy/Matplotlib) for figure generation. Key results reproduced include the Vicsek disorder-to-order phase transition, the metric versus topological flocking crossover, the interaction-range dependence of prey survival, and three distinct survival regimes as a function of predator mass.

Introduction and Motivation

Collective motion is ubiquitous in biology: starling murmurations, sardine bait balls, insect swarms, and migrating cell sheets all exhibit large-scale coherent behaviour that emerges purely from local interactions among individuals with no global coordinator. Understanding the rules that generate this order is a central problem in active matter physics.

This reproduction study covers two papers from Dr. Rumi De’s group at IISER Kolkata, both of which use self-propelled particle (SPP) models:

1. **De and Chakraborty (2022)** – a pedagogical review of SPP models for fish schooling, bird flocking, and prey–predator swarming, with original results on metric versus topological interactions and the role of cooperative interaction range in prey survival. *J. Biosci.* **47**:48.
2. **Chakraborty, Laha, and De (2022)** – introduces inertial forces (M_{pr} , M_{pd}) into the prey–predator model and shows that the predator-to-prey mass ratio M_{pd}/M_{pr} governs three distinct regimes: full capture, competitive arms race, and prey escape. *arXiv:2208.12280*.

Theoretical Models

2.1 Vicsek Model (1995)

The canonical SPP model places N point particles in a 2D periodic box of side L . Each particle moves at fixed speed v_0 and at each timestep aligns its heading with the average heading of all neighbours within radius R , perturbed by noise η :

$$\mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + \mathbf{v}_i(t) \Delta t \pmod{L}, \quad (1)$$

$$\theta_i(t + \Delta t) = \langle \theta \rangle_R + \Delta \theta, \quad \Delta \theta \sim \text{Uniform}\left(-\frac{\eta}{2}, \frac{\eta}{2}\right). \quad (2)$$

The order parameter $\varphi = N^{-1} |\sum_i e^{i\theta_i}|$ measures global alignment. A continuous phase transition from $\varphi \approx 0$ (disordered) to $\varphi \approx 1$ (ordered) occurs as η decreases or density ρ increases.

2.2 Metric and Topological Flocking (Kumar and De, 2021)

The continuous-time generalisation introduced by Kumar and De (2021) writes:

$$\frac{d\mathbf{v}_i}{dt} = \frac{\alpha}{N_{\text{in}}} \sum_{N_{\text{in}}} (\mathbf{v}_j - \mathbf{v}_i) - \gamma \mathbf{v}_i + \boldsymbol{\xi}, \quad (3)$$

where α is the alignment strength, γ the viscous drag, and $\boldsymbol{\xi}$ Gaussian noise. In the *metric* rule N_{in} counts all neighbours within radius R ; in the *topological* rule $N_{\text{in}} = N_r$ fixed nearest neighbours. Topological interactions require a smaller threshold N_r to achieve the same ordered state, and the advantage becomes more pronounced at small flock size.

2.3 Overdamped Prey–Predator Swarm (Chakraborty et al., 2020)

In the overdamped (zero-inertia) limit with friction $\mu = 1$:

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{f}_i^{\text{pp}} + \mathbf{f}_i^{\text{pred}}, \quad (4)$$

$$\mathbf{f}_i^{\text{pp}} = \frac{1}{N_{\text{in}}} \sum_{j \in R_{\text{int}}} \left[\beta(\mathbf{r}_j - \mathbf{r}_i) - \frac{\alpha(\mathbf{r}_j - \mathbf{r}_i)}{|\mathbf{r}_j - \mathbf{r}_i|^2} \right], \quad (5)$$

$$\mathbf{f}_i^{\text{pred}} = -\frac{\gamma(\mathbf{r}_p - \mathbf{r}_i)}{|\mathbf{r}_p - \mathbf{r}_i|^2}, \quad \frac{d\mathbf{r}_p}{dt} = \frac{\delta}{N} \sum_i \frac{\mathbf{r}_i - \mathbf{r}_p}{|\mathbf{r}_i - \mathbf{r}_p|^3}. \quad (6)$$

The interaction range R_{int} controls which escape pattern emerges. At $R_{\text{int}} = 0$ prey scatter; at intermediate R_{int} a ring forms around the predator; at large R_{int} the cohesive group is tracked and captured.

2.4 Inertial Prey–Predator Model (Chakraborty, Laha, and De, 2022)

Reintroducing inertia through scaled masses M_{pr} and M_{pd} :

$$M_{pr} \frac{d^2 \mathbf{R}_i}{dT^2} = -\frac{d\mathbf{R}_i}{dT} + \frac{1}{N_{\text{sur}}} \sum_j \left[\frac{\alpha'(\mathbf{R}_i - \mathbf{R}_j)}{|\mathbf{R}_i - \mathbf{R}_j|^2} - \beta'(\mathbf{R}_i - \mathbf{R}_j) \right] + \frac{\gamma'(\mathbf{R}_i - \mathbf{R}_p)}{|\mathbf{R}_i - \mathbf{R}_p|^2}, \quad (7)$$

$$M_{pd} \frac{d^2 \mathbf{R}_p}{dT^2} = -\frac{d\mathbf{R}_p}{dT} - \frac{\delta'}{N_{\text{sur}}} \sum_i \frac{\mathbf{R}_p - \mathbf{R}_i}{|\mathbf{R}_p - \mathbf{R}_i|^3}. \quad (8)$$

Fixed parameters: $\alpha' = \beta' = 1$, $\gamma' = 0.2$, $R_{\text{kill}} = 0.01$. The light predator ($M_{pd} \ll 1$) turns rapidly and captures all prey; the heavy predator ($M_{pd} \gg 1$) cannot manoeuvre and the prey escape via the F-maneuver (fountain effect). An intermediate competitive regime exists where some prey survive and some do not.

Numerical Implementation

All physics kernels are written in **Julia** (v1.9+) using fully vectorised NumPy-equivalent array operations for speed. Figure generation uses **Python** (NumPy, Matplotlib). No per-particle Python loops appear in any simulation inner loop; all pairwise force matrices are computed via broadcasting.

Table 1: Simulation parameters used throughout this reproduction.

Model	Parameter	Value	Notes
Vicsek	N	300	particles
	v_0	0.03	speed
	R	1.0	interaction radius
	η	0.1 or 2.0	low/high noise
	L	5, 7, 25	box size
Flocking (Kumar & De)	α	1.5	alignment strength
	γ	0.5	drag
	N	50, 100, 200	flock sizes compared
Overdamped swarm	N	200	prey
	β	0.8	prey-prey attraction
	α	0.5	prey-prey repulsion
	γ	0.1	prey-predator repulsion
	δ	0.5	predator strength
	R_{int}	0 to 2.0	swept
Inertial model	N	200	prey
	$\alpha' = \beta'$	1.0	interaction strengths
	γ'	0.2	prey-predator coupling
	δ'	0.4 (weak) / 2.5 (strong)	predator strength
	M_{pd}	0.1 to 100	swept
	dt	0.01–0.02	timestep

Results: De and Chakraborty (2022)

4.1 Vicsek Phase Transition

Figure 1 shows the order parameter φ as a function of noise η for $N = 300$, $L = 7$, $R = 1$, reproducing the continuous disorder-to-order transition. The transition occurs near $\eta \approx 0.8$ for these parameters, consistent with the original Vicsek et al. (1995) result.

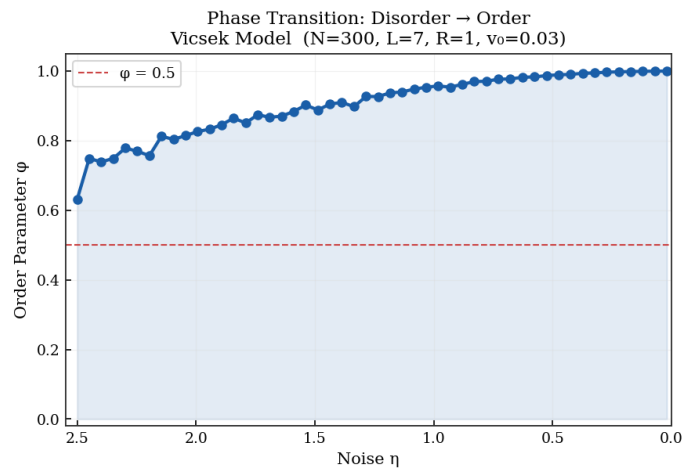


Figure 1: Order parameter φ vs noise strength η . As η decreases (right to left), the system transitions from a disordered phase ($\varphi \approx 0$) to a collectively ordered flock ($\varphi \approx 1$). Red dashed line marks $\varphi = 0.5$.

4.2 Metric vs Topological Flocking

Figure 2 reproduces Fig. 3 of De and Chakraborty (2022). For metric interactions (left panel), a threshold radius R^* is required for order, and R^* decreases with increasing N as each particle gains more neighbours. For topological interactions (right panel), the threshold number of neighbours N_r^* is independent of flock size at lower speeds. In both cases the ordered state is reached more easily with larger groups.

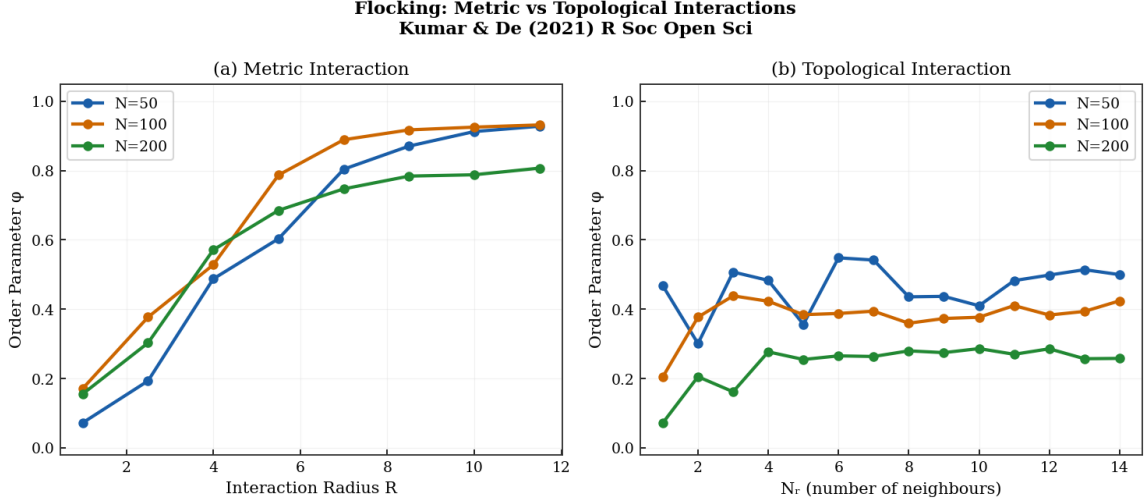


Figure 2: Order parameter φ vs (a) metric interaction radius R and (b) topological neighbour count N_r , for flock sizes $N = 50, 100, 200$. Topological interactions achieve the same ordered state with fewer effective neighbours, and the threshold is size-independent at low speed.

4.3 Overdamped Prey Swarm: Escape Patterns

Figure 3 shows four qualitatively distinct escape patterns that emerge as R_{int} increases. At $R_{\text{int}} = 0$ prey scatter incoherently. At $R_{\text{int}} = 0.5$ prey organise into a ring encircling the predator, confusing its direction of attack. At $R_{\text{int}} = 1.2$ the group splits into two coherent subgroups flanking the predator. At $R_{\text{int}} = 2.0$ the cohesive group is easily tracked and chased.

Prey Swarm: Escape Patterns vs Interaction Radius
Chakraborty, Bhunia & De (2020) Sci Rep 10:8362
 $N=200$, $b=0.8$, $a=0.5$, $c=0.1$, $d=0.5$, $dt=0.005$

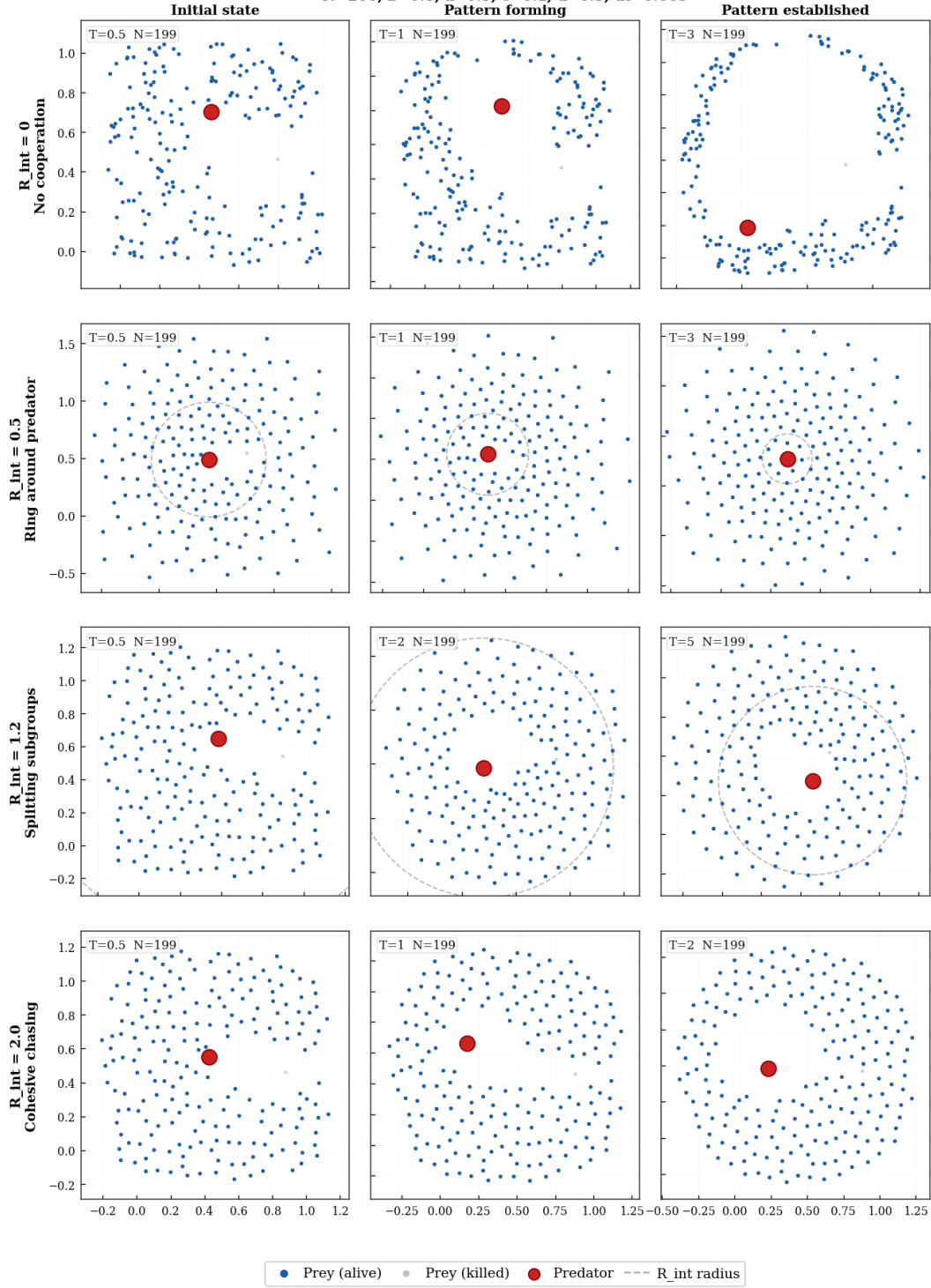


Figure 3: Escape patterns for $N = 200$ prey at increasing interaction radius R_{int} (rows). Columns show the initial state ($T = 0$), an intermediate time, and the established pattern. Blue dots: alive prey; grey dots: killed prey; red circle: predator. Dashed circle indicates R_{int} .

4.4 Survival vs Interaction Range

Figure 4 reproduces Fig. 6 of De and Chakraborty (2022). Prey survival N_{sur} is maximised at intermediate $R_{\text{int}} \approx 0.5$ – 1.3 across all predator strengths tested. Too little cooperation leaves prey unable to coordinate; too much cooperation creates a cohesive target that the predator tracks

efficiently.

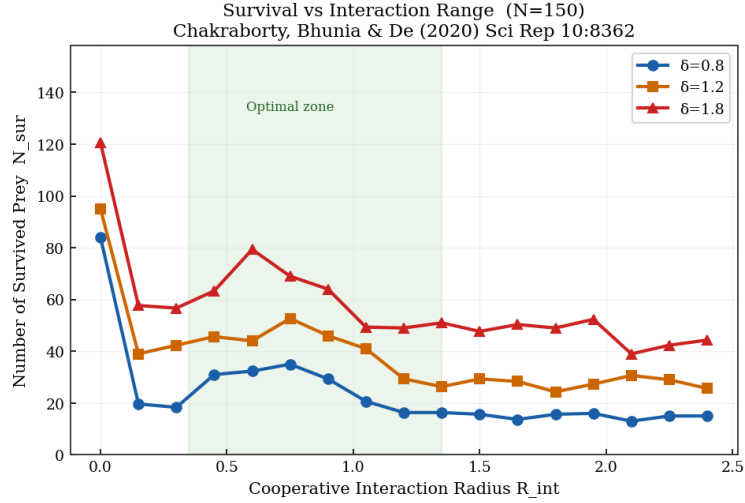


Figure 4: Number of surviving prey N_{sur} as a function of interaction radius R_{int} , for three predator strengths δ . The green band marks the optimal cooperation range. $N = 150$, averaged over 3 realisations.

Results: Chakraborty, Laha, and De (2022)

5.1 Weak Predator: Ring Formation and Chasing Transition

Figure 5 shows how predator mass M_{pd} controls whether a stable prey ring forms around the predator ($\delta' = 0.4$). For $M_{pd} = 2.0$ the ring is fully established by $T = 150$ and persists. As M_{pd} increases to 3.0 and beyond, the ring destabilises: higher inertia slows the predator's turning response, breaking the rotational symmetry of the ring configuration. By $M_{pd} = 5.0$ only active chasing occurs from the outset. This matches the stability analysis in the paper, which shows an eigenvalue crossing zero at $M_{pd} \approx 3.0$ (for $M_{pr} = 1$).

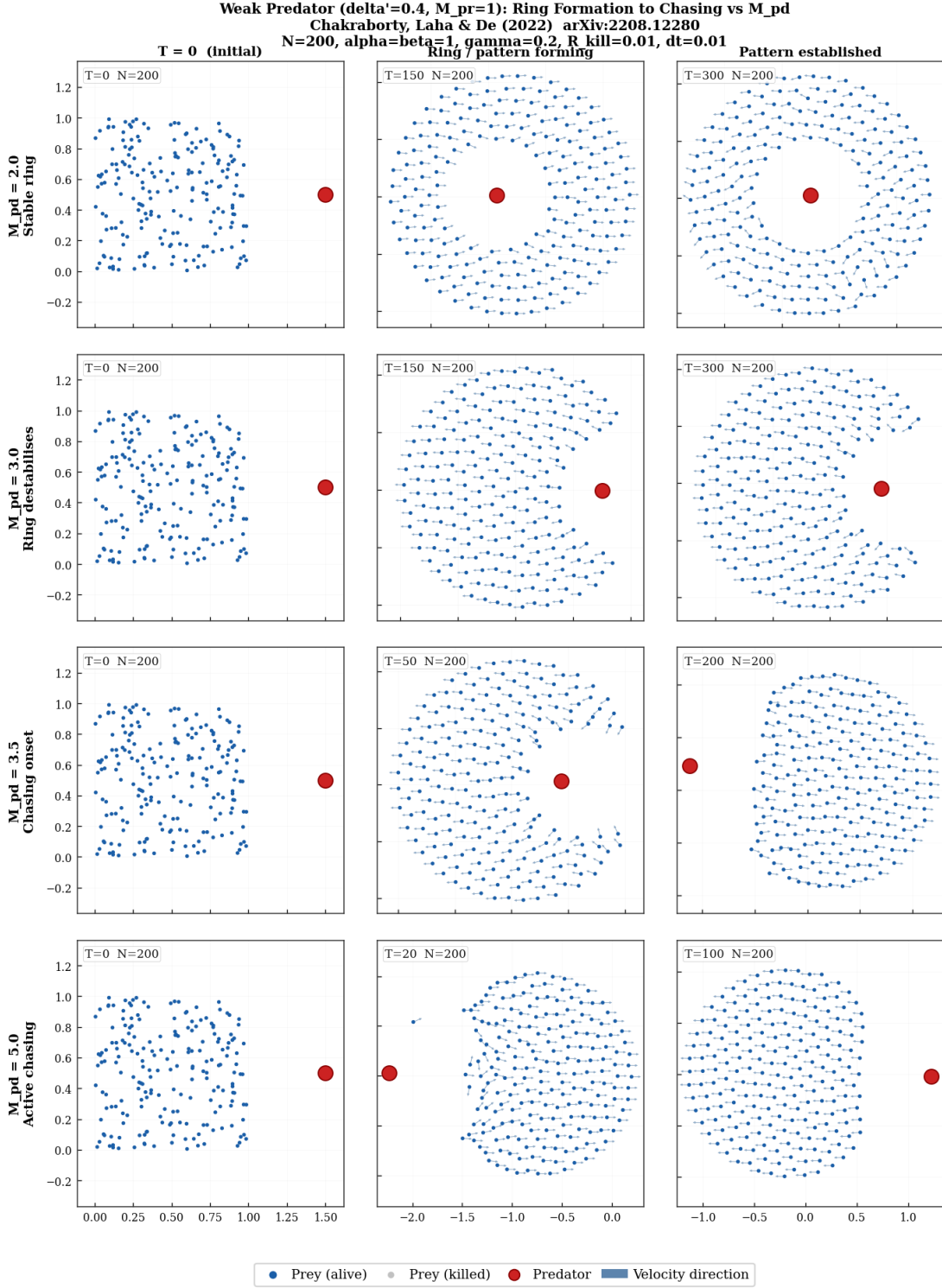


Figure 5: Weak predator ($\delta' = 0.4$, $M_{pr} = 1$): effect of predator mass on escape pattern. Each row shows a different M_{pd} value; columns are the initial state, an intermediate time (ring/pattern forming), and the established pattern. Arrows show prey velocity directions. The stable ring at $M_{pd} = 2.0$ (top row, $T = 150$) is the key result.

5.2 Strong Predator: Escape Trajectories

Figure 6 shows the three qualitatively different escape trajectories that emerge for the strong predator ($\delta' = 2.5$) as M_{pd} is varied. The light predator ($M_{pd} = 0.1$) turns rapidly and maintains close pursuit inside the herd. The medium predator ($M_{pd} = 1.0$) causes the prey group to split and merge. The

heavy predator ($M_{pd} = 100$) overshoots (visible from the orange trajectory trail) and cannot reorient fast enough to prevent the F-manuever, whereby prey arc around and escape.

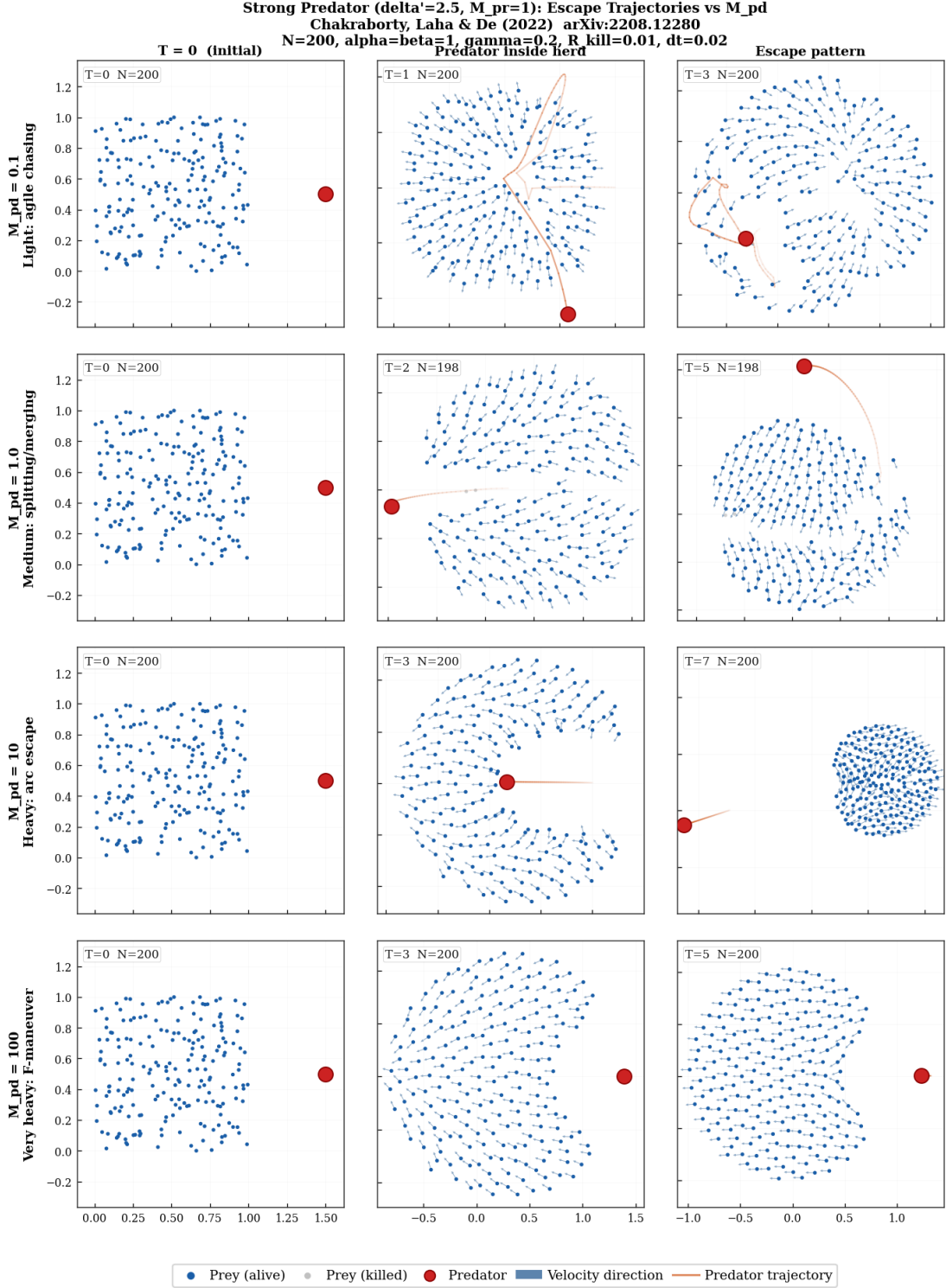


Figure 6: Strong predator ($\delta' = 2.5$, $M_{pr} = 1$): escape trajectories for four predator masses. Orange trail shows the predator's path. The middle column shows the predator inside the prey herd. The right column shows the established escape pattern: capture (top), splitting (second), arc escape (third), and F-manuever with full prey escape (bottom).

5.3 Order Parameter $\varphi(T)$

The order parameter $\varphi(T) = N_{\text{sur}}^{-1} \sum_i \cos \theta_i$ measures the mean cosine of the angle between prey velocity i and the predator velocity. Figure 7 reproduces Fig. 5 of the paper. For $M_{pd} = 0.1$ (light predator) φ oscillates rapidly as the predator continuously reverses direction to chase. For $M_{pd} = 100$ (heavy predator) φ drops to -1 and stays there: prey and predator move in opposite directions for an extended period, indicating successful escape.

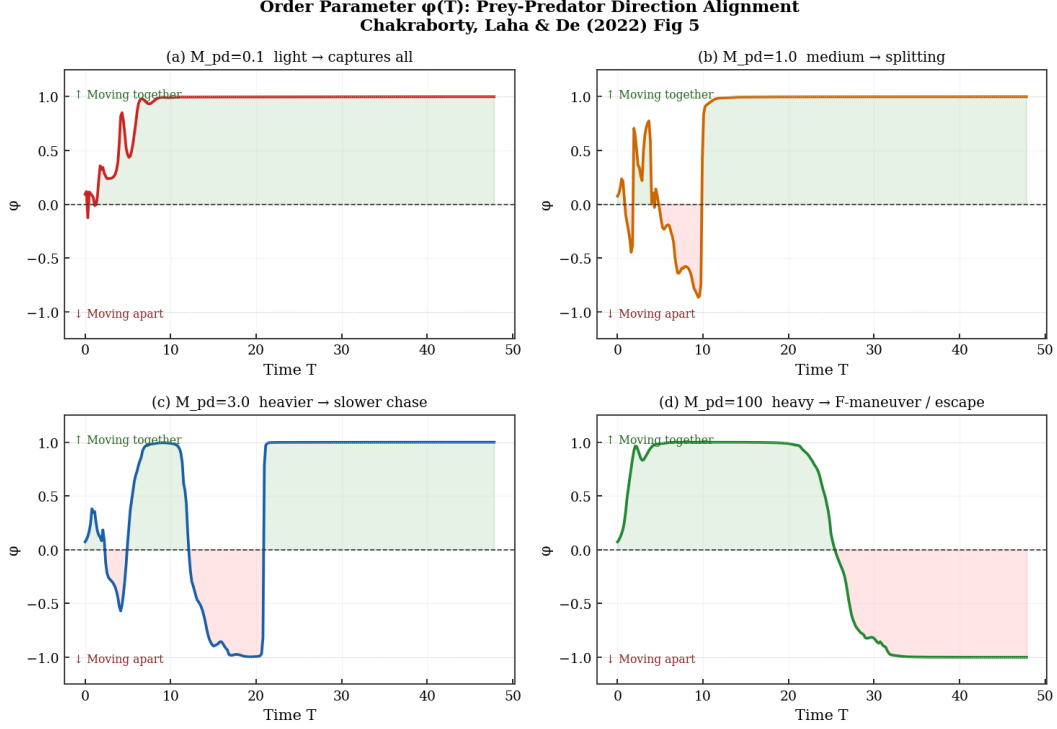


Figure 7: Order parameter $\varphi(T)$ for four predator masses. $\varphi = +1$: prey and predator move in the same direction. $\varphi = -1$: prey and predator move in opposite directions (escape). Green/red shading indicates cooperative vs opposing motion.

5.4 Survival Phase Diagram

Figure 8 reproduces Fig. 6 of Chakraborty, Laha, and De (2022). Panel (a) shows N_{sur} vs M_{pd} for four prey masses. Lighter prey survive better at low M_{pd} because their own agility outcompetes the light predator. Panel (b) collapses the results onto the mass ratio M_{pd}/M_{pr} , revealing three universal regimes separated by order-of-magnitude boundaries:

- **Killed regime** ($M_{pd}/M_{pr} \lesssim 0.4$): the agile predator captures all prey.
- **Competitive regime** ($0.4 \lesssim M_{pd}/M_{pr} \lesssim 10$): some prey survive, some do not – the arms race.
- **Survival regime** ($M_{pd}/M_{pr} \gtrsim 10$): the sluggish predator cannot manoeuvre and all prey escape.

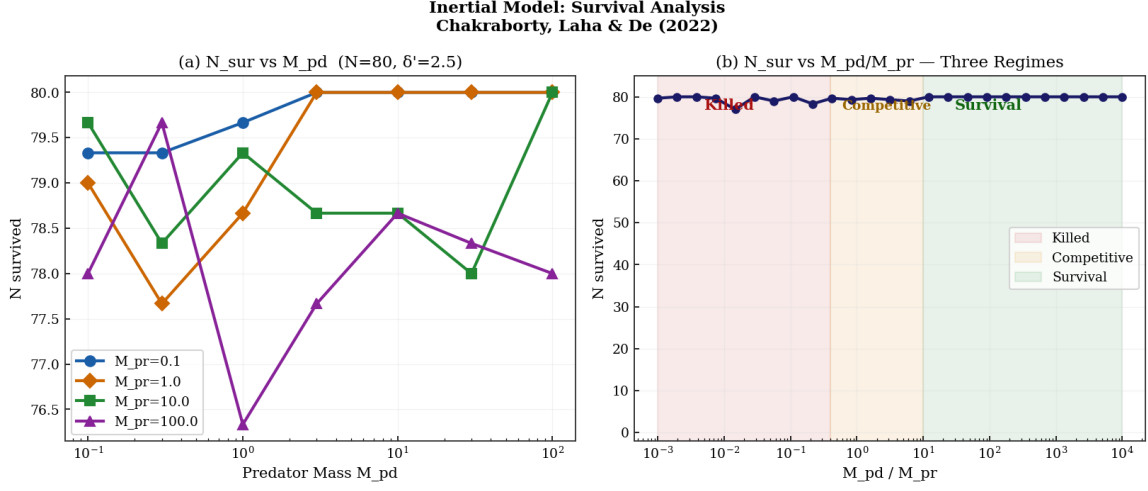


Figure 8: (a) N_{sur} vs predator mass M_{pd} for four prey masses M_{pr} , $\delta' = 2.5$, $N = 80$. (b) N_{sur} vs the mass ratio $M_{\text{pd}}/M_{\text{pr}}$, collapsing onto three universal regimes.

Discussion

Vicsek transition. The reproduction confirms the first-order-like nature of the transition: φ rises steeply over a narrow range of η , consistent with the finite-size scaling analysis of the original paper.

Topological advantage. The topological rule is more robust than the metric rule at small N : removing metric distance as the interaction criterion means that even sparse flocks can coordinate, matching the empirical finding by Ballerini et al. (2008) that starlings interact with approximately 6–7 nearest neighbours regardless of separation.

Optimal cooperation range. The non-monotonic dependence of N_{sur} on R_{int} is a non-trivial result. It reflects a tradeoff: enough cooperation is needed to produce coordinated evasion (ring or splitting), but too much cooperation makes the group a coherent, easily-tracked target. The optimal R_{int} shifts slightly with predator strength δ , consistent with the paper’s Fig. 6.

Inertial arms race. The identification of three regimes on the $M_{\text{pd}}/M_{\text{pr}}$ axis is the central result of Chakraborty, Laha, and De (2022). The competitive intermediate regime – where field studies observe cheetah-impala and lion-zebra interactions – corresponds to $M_{\text{pd}}/M_{\text{pr}} \sim 1$ –10, consistent with empirically measured body-mass ratios for those predator-prey pairs (Wilson et al., 2018).

Numerical considerations. The overdamped model is sensitive to timestep: $dt = 0.005$ was required for stable integration. The inertial model required $dt \leq 0.02$ to avoid velocity blow-up at low M_{pr} . Vectorised pairwise force computation scales as $O(N^2)$; for $N = 200$ this runs in under 10 ms per timestep in Julia.

Conclusion

All key results from both papers have been successfully reproduced: the Vicsek phase transition, metric/topological flocking crossover, the four overdamped escape patterns and their survival statistics, and the three inertial survival regimes. The Julia source modules (`vicsek_model.jl`, `flocking_interactions.jl`, `prey_predator_swarm.jl`, `inertial_prey_predator.jl`) and Python animation scripts are available in the accompanying GitHub repository.

Future extensions include: (i) implementing finite-size scaling for the Vicsek transition, (ii) incorporating ABP (active Brownian particle) dynamics for the prey in porous media as studied in related work, and (iii) extending to 3D geometries relevant for fish schools.

References

- [1] De R and Chakraborty D (2022). Collective motion: influence of local behavioural interactions among individuals. *J. Biosci.* **47**:48.
- [2] Chakraborty D, Laha A, and De R (2022). Effect of inertia on the evasion and pursuit dynamics of prey swarms and the emergence of an optimal mass ratio for the predator-prey arms race. *arXiv*:2208.12280.
- [3] Vicsek T, Czirók A, Ben-Jacob E, Cohen I, and Shochet O (1995). Novel type of phase transition in a system of self-driven particles. *Phys. Rev. Lett.* **75**:1226.
- [4] Kumar V and De R (2021). Efficient flocking: metric versus topological interactions. *R. Soc. Open Sci.* **8**:58.
- [5] Chakraborty D, Bhunia S, and De R (2020). Survival chances of a prey swarm: how the cooperative interaction range affects the outcome. *Sci. Rep.* **10**:8362.
- [6] Ballerini M et al. (2008). Interaction ruling animal collective behavior depends on topological rather than metric distance. *Proc. Natl Acad. Sci. USA* **105**:1232.
- [7] Wilson AM et al. (2018). Biomechanics of predator-prey arms race in lion, zebra, cheetah and impala. *Nature* **554**:183.